PROJECT REPORT · OCTOBER 2026 ~

Smart Fake News Detection System

A text classification project using TF-IDF feature representations and linear models. Headline benchmark score: 97.59% accuracy.

AUTHOR & COHORT

Aditya Rathore

AIML - A6/SEP-11772

AI/ML with Python Training Program
September 2026 batch

SUPERVISION & INSTITUTION

Pratham Saxena

Trainer & Project Guide

InternsElite Academic Assessment
Submitted October 2026

Responsible Use Notice: This system is an educational prototype and research baseline. Learned linguistic patterns are statistical associations, not verified fact-checks. Classification outputs do not constitute proof of veracity or falsity.

Repository: github.com/AdiRathoreX/Fake_news_detection_classifier · InternsElite AIML Program

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

EXECUTIVE SUMMARY ~

A strong benchmark score, not a fact checker

This investigation examines whether statistical n-gram patterns in combined news headlines and article bodies can separate labeled classes in the public WELFake benchmark. Using standardized TF-IDF vectorization and supervised linear algorithms, three classical models were trained on an 80% split and tested against a held-out evaluation corpus of 14,427 articles. Linear SVM performed best at 97.59% accuracy; this benchmark result does not verify whether real-world news is true or false.

Model	Accuracy	Macro F1
Naive Bayes	86.8%	86.8%
Logistic Regression	95.2%	95.2%
Linear SVM (final)	**97.59%**	**97.58%**

97.59%
HELD-OUT ACCURACY

14,427
TEST OBSERVATIONS

72,134
TOTAL DATASET RECORDS

While the final Linear Support Vector Machine demonstrates high classification fidelity within the closed test split, this metric does not establish external fact-checking efficacy. Models exposed to uncurated current events, shifting vocabularies, or novel geopolitical narratives will produce erroneous inferences without retrieval verification.

Source: Project experiment logs, Aditya Rathore (September–October 2026).

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

INTRODUCTION & PROBLEM STATEMENT ~

Purpose, scope and objectives

This project was developed as an educational inquiry into natural language processing workflows, centering on binary classification mechanics, sparse feature spaces, and disciplined model validation. The primary research question addresses whether surface textual signals alone provide sufficient separation between dataset classes. This project tests whether TF-IDF features and supervised models can distinguish the dataset's two labels from article text. It does not determine whether a story is factually true.

SYSTEM OBJECTIVES

- Inspect and profile a large-scale corpus of news titles, article bodies, and binary labels.
- Construct clean text inputs by stripping whitespace and concatenating titles and bodies.
- Train and contrast Multinomial Naive Bayes, Logistic Regression, and Linear SVM.
- Assess generalization through held-out test splits and export artifacts for deployment.
- Design an interactive Streamlit user interface communicating uncertainty boundaries.

IN SCOPE VS. ESTABLISHED

In Scope: Text-only pre-processing, TF-IDF n-gram extraction, offline model benchmarking, confusion matrix diagnostics, and local web deployment.

Not Established: Factual claim verification, source reputation scoring, knowledge graph grounding, resilience to distribution drift, or multilingual adaptation.

Epistemic Boundary: "Likely fake" and "Likely real" are the model's predicted labels, based on text patterns learned during training. LinearSVC does not produce calibrated class probabilities. These labels do not prove a story true or false.

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

DATASET ANALYSIS & ANOMALY REPORTING ~

The data and a label-count discrepancy

The project utilizes the WELFake dataset comprising 72,134 records across four initial fields: an index column, title, text body, and binary target label. Exploratory data audits identified 558 missing titles and 39 missing body strings, which were imputed with empty strings. Zero duplicate rows were found.

AUDITED DATASET TOTALS

Within the downloaded CSV, raw label distribution resolves to:

- **Label 0:** 35,028 rows (48.56%)
- **Label 1:** 37,106 rows (51.44%)

A significant metadata discrepancy exists in the upstream source documentation [1]. While the textual description defines Label 0 as Fake and Label 1 as Real, the accompanying source documentation associates those exact numeric totals with the inverse class names. The underlying mapping remains unverified in external releases.

CHALLENGE & RESPONSE

Challenge: The dataset source materials conflict about which class names match the reported label totals.

Response: This report gives the audited counts as Label 0 and Label 1, flags the mapping for verification, and retains the notebook's [Fake, Real] convention for model metrics.

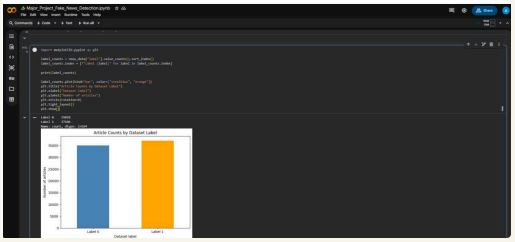


Figure 1: Extracted article count distribution from project notebook showing raw numeric binary labels.

Source: WELFake dataset audit; Verma et al., Zenodo (2021) [1].

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

FEATURE ENGINEERING & CORPUS PROFILE ~

Preparation and exploration

Data cleaning followed a deterministic transformation sequence to produce consistent textual records for sparse linear tokenization. The redundant row index column was removed, missing values were resolved, and document components were merged.

TRANSFORMATION STEPS

1. **Index Elimination:** Dropped the redundant `Unnamed: 0` column.
2. **Missing Imputation:** Replaced 558 null titles and 39 null texts with empty strings.
3. **Whitespace Normalization:** Stripped leading and trailing whitespace characters.
4. **Document Fusion:** Concatenated title and body into a single `content` field.
5. **Integrity Verification:** Confirmed all 72,134 rows contain non-empty combined strings.

CORPUS WORD COUNT PROFILE

552.7

MEAN WORD COUNT PER DOCUMENT

Text length across the WELFake corpus exhibits substantial right-skewness. While short articles and briefs center around 200–400 words, detailed investigative entries and transcripts extend well past 1,500 words.

The distribution tail reaches several thousand words, necessitating robust document-level normalization during term-frequency matrix construction.

72,134
Total Cleaned Rows

0
Empty Combined Records

1,500+
Histogram Bound (Words)

Source: WELFake exploratory data analysis notebook, InternsElite project repository.

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

EXPERIMENTAL METHODOLOGY ~

Method: a reproducible modeling workflow

The modeling pipeline implements an empirical split-transform-fit routine using scikit-learn. To prevent data contamination, dataset splitting precedes feature extraction, and TF-IDF parameters were selected to balance computational tractability with n-gram expressive capacity.

Step	Operation	Project detail
01 Prepare	Join title and body	Concatenate normalized title and text strings into a unified content vector.
02 Split	Train / test partitioning	80/20 stratified split (57,707 train / 14,427 test); seeded at `random_state=42`.
03 Vectorize	TF-IDF feature extraction	Final configuration: `max_features=50,000`, unigrams & bigrams (1,2), `min_df=2`, English stop words removed, and `sublinear_tf=True`.
04 Fit	Linear SVM optimization	Train `LinearSVC` with regularization parameter `C=1.0` over the sparse document-term matrix.
05 Evaluate	Held-out metrics	Calculate accuracy, macro F1 score, confusion matrices, and per-class precision/recall metrics.
06 Deploy	Streamlit packaging	Save the selected Linear SVM model and fitted TF-IDF vectorizer as Joblib files. The Streamlit app loads these files to make predictions.

Mathematical feature weighting applies standard sublinear TF-IDF scaling where term frequency is dampened by $1 + \log(\mathrm{TF})$ and scaled by inverse document frequency $\mathrm{IDF}(t) = \log\left(\frac{1+n}{1+\mathrm{DF}(t)}\right) + 1$, preventing high-frequency tokens from dominating the linear decision boundary [3, 4].

Source: Implementation workflow adapted from scikit-learn pipeline documentation [3, 4].

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

COMPARATIVE EVALUATION ~

Results: three models, one held-out set

All performance metrics were generated from the unobserved held-out partition of 14,427 articles under identical 80/20 stratified conditions. An initial experiment using a 20,000-feature ceiling produced an SVM accuracy of approximately 96.64%; expanding vocabulary capacity to 50,000 bigrams yielded the final candidate reaching 97.59% accuracy.

Model	Test accuracy	Correct predictions	Role
Multinomial Naive Bayes	86.8%	12,529 / 14,427	Probabilistic baseline
Logistic Regression	95.2%	13,733 / 14,427	Linear comparison
Linear SVM (final)	**97.59%**	**14,079 / 14,427**	**Selected for app**

CONFUSION MATRICES

Rows are actual labels [Fake, Real]. Within each row, counts are predicted labels [Fake, Real].

Naive Bayes: [5,925, 1,081] / [817, 6,604]
Logistic Regression: [6,603, 403] / [291, 7,130]
Linear SVM: [6,789, 217] / [131, 7,290]

MACRO METRICS & COMPARISON

Macro precision / recall / F1

· Naive Bayes: 86.91% / 86.78% / 86.81%

· Logistic Regression: 95.21% / 95.16% / 95.18%

·Linear SVM: 97.61% / 97.57% / 97.58%

Naive Bayes is a probabilistic baseline with an independence assumption. Logistic Regression performs better; Linear SVM scores highest on this held-out split.

Source: Final model evaluation cell in the project notebook.

Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

ERROR ANALYSIS & CLASS DIAGNOSTICS ~

Final model: Linear SVM detailed performance and errors

A rigorous examination of the selected Linear Support Vector Machine reveals balanced performance across classes, alongside subtle asymmetries in false-positive and false-negative distributions. Confusion matrix rows represent ground truth; columns indicate model predictions ordered [Fake, Real] [5].

CONFUSION MATRIX (HELD-OUT TEST SET)

Actual class	Pred. Fake	Pred. Real	Support
Fake	6,789	217	7,006
Real	131	7,290	7,421

PER-CLASS VALIDATION METRICS

Class	Precision	Recall	F1 score
Fake	0.981	0.969	0.975
Real	0.971	0.982	0.977
Macro avg	**0.976**	**0.976**	**0.976**

Error Dynamics: Exactly 217 Fake-labeled stories were erroneously predicted as Real, while 131 Real-labeled stories were flagged as Fake. Consequently, Fake recall (0.969) is lower than Real recall (0.982). Because the dataset metadata carries an unresolved label-count discrepancy, real-world error costs cannot be definitively assigned without resolving source semantics.

Source: Scikit-learn classification report and confusion matrix evaluation [5].

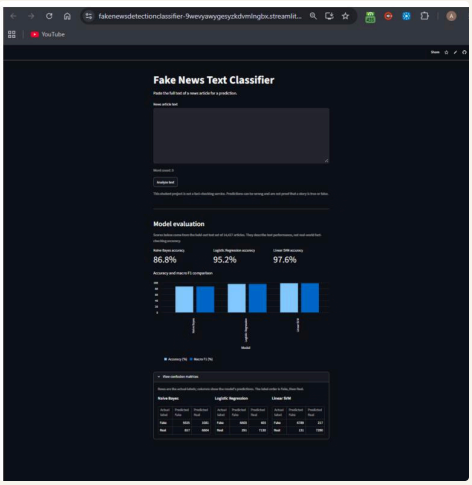
Made with GAMMA

SMART FAKE NEWS DETECTION SYSTEM

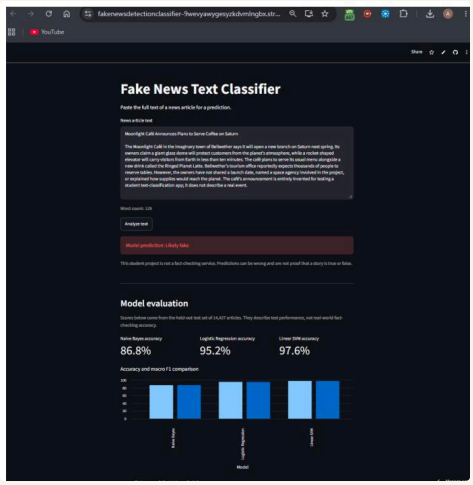
INTERACTIVE DEPLOYMENT ~

The Streamlit demo application

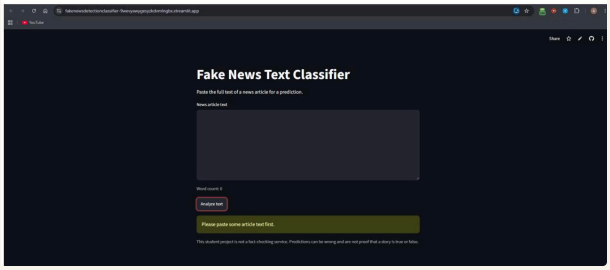
The app loads the saved Linear SVM and TF-IDF vectorizer and returns a likely-label message. It shows the model comparison and confusion matrices as held-out test context, not real-world accuracy.



Dashboard: dashboard.



Prediction: prediction on fictional Moonlight Café on Saturn text ("Likely fake", demonstrates app behavior, not validation).



Empty-input warning: empty-input warning.

Operational guardrails: The app warns on empty input and requires at least 100 words; a guardrail, not a reliability guarantee.

Source: Streamlit user application screenshots, Aditya Rathore (app.py).

Made with GAMMA

CRITICAL APPRAISAL & FUTURE WORK ~

Limitations, next steps, conclusion and references

LIMITATIONS

- Dataset bias; potential leakage or shortcuts; distribution shift; label uncertainty; no evidence retrieval; short-input risk.
- Not a fact-checking service: text patterns alone do not establish truth, intent or provenance.

NEXT STEPS

1. **Critical:** Resolve label mapping.
2. **High:** Pipeline and cross-validation.
3. **High:** Check duplicates and near-duplicates across train/test.
4. **High:** Time- or source-separated evaluation.
5. **Medium:** Calibration, per-class metrics, confidence limits.
6. **Medium:** Evidence links and human review.

CONCLUSION

End-to-end workflow. The final Linear SVM reaches **97.59% accuracy** on the held-out test set. Reconcile label metadata and test on independent data before stronger claims.

REFERENCES

[1] S. K. Verma, A. K. Yadav, A. R. Sharma, "WELFake: A Large-Scale Dataset for Detecting Fake News," Zenodo, 2021. https://zenodo.org/records/4561253

[2] S. K. Verma et al., "WELFake: a large-scale dataset for detecting fake news," IEEE Trans. Computational Social Systems. DOI 10.1109/TCSS.2021.3068519

[3] scikit-learn, TfidfVectorizer documentation; [4] LinearSVC documentation; [5] confusion_matrix documentation

[6] Project source materials by Aditya Rathore: notebook, app.py, README.md, requirements.txt, model artifacts, screenshots

Repository: github.com/AdiRathoreX/Fake_news_detection_classifier

Made with GAMMA